Excluding transitive-side equality under an alternating tail

Human Compression project: bounded L044 proof component

Let $G \le A_n$, $n \ge 12$, have a transitive string C-marking with all maximal omissions intransitive. Suppose a unique eligible interior i-edge ab joins the two omission orbits A, B, and $L = G_{<i}$ fixes B pointwise and is transitive on A. Assume the equality configuration of the accepted transitive-side support theorem: with $h = i + 1 < r - 1$ and $K = G_{>h}$,

$$|A| = 2h, \qquad \mathrm{Fix}(K) = A \cup \{b\}.$$

Put $X = B \setminus \{b\}$, and suppose K is the full natural alternating group A_X. Write $m = h = i + 1 \ge 2$ and $\tau = \rho_{i+1}$.

Proposition 1. *The stated equality configuration is impossible.*

Proof. The accepted transitive-side proof shows that τ is fixed-point-free on A and centralizes L. Its pairs form an L-invariant system of m blocks. Every distinguished generator of L has essential block image, by the actual global-omission argument in that proof. These $m-1$ images are independent, so Whiston's equality clause makes the block image the full S_m.

Write $\lambda = \tau|_A$, $a' = a^\lambda$, and $H = G_{<i-1}$. The generators of H fix a by unique crossing and fix a' by commutation with τ. Their $m - 2$ independent block images fix the pair $\{a, a'\}$ and generate the full S_{m-1} on the others, again by Whiston. For $m = 2$ this is the trivial S_1 statement. No classification of a maximal symmetric string and no projected IP are used.

Since the nonempty involutory family generating $K = A_X$ cannot generate A_3 or A_4, one has $|X| \ge 5$. The crossing generator $s = \rho_i$ centralizes K, preserves its unique nontrivial orbit X, and fixes X pointwise because the natural alternating centralizer is trivial. Consequently

$$s = (a\ b)u,$$

where u is supported on $A \setminus \{a\}$ and centralizes H. For $m \ge 3$ the only H-fixed points in A are a, a', so u fixes a' as well.

First suppose m is even. The restriction $\tau|_B$ is even and moves b into X. The natural alternating groups on $B \setminus \{b\}$ and $B \setminus \{b^\tau\}$ generate A_B, by the elementary overlapping-support three-cycle argument. Thus $R = G_{>i} = \langle \tau, K \rangle$ is a subgroup of $\langle \lambda \rangle \times A_B$ projecting onto both factors. Perfection of A_B implies $\lambda \in R$: otherwise the projection to A_B would identify R with a graph of a nontrivial homomorphism $A_B \to C_2$. Since L and R have disjoint index sets, IP gives $\lambda \notin L$.

Identify the pair stabilizer with $\mathbb{F}_2^m \rtimes S_m$. Its base vectors record the within-pair flips; block permutations are even on the $2m$ points. Hence the kernel W of the L block action is contained in the even-weight space E and is invariant under S_m. If W contains a nonzero constant vector, it contains λ. If it contains a nonconstant vector, its difference with a transposition-conjugate having unequal coordinates has weight two. Its S_m-conjugates generate E. For even m, $\lambda \in E$, again contradicting $\lambda \notin L$. Thus $W = 0$ and $L \cong S_m$. But H fixes a and has order $(m-1)!$, since its block image is S_{m-1}. Transitivity of L on $2m$ points would instead require

$$|L_a| = \frac{m!}{2m} = \frac{(m-1)!}{2},$$

which cannot contain H. This excludes all even m.

For odd $m \geq 5$, apply the accepted paired-row centralizer lemma to H on the other $m-1$ pairs. Their number is even and at least four, and the block image is S_{m-1}. Every centralizing involution there is even. Since u fixes a, a', it is even on A, making $(a\ b)u$ odd. This contradicts $G \leq A_n$.

The only remaining value is $m = 3$. The two generators of L belong to $E \rtimes S_3$, since they are even on the six points, and their block images are distinct transpositions generating S_3. Their product has a three-cycle block image. For any even vector $v \in \mathbb{F}_2^3$ and a three-cycle π, the cube of (v, π) has base vector

$$v + \pi v + \pi^2 v = \Big(\sum_j v_j\Big)(1,1,1) = 0.$$

Thus the product of the two involutions has order three. They generate a group of order at most six; the surjective block image shows $L \cong S_3$ exactly. Yet H is a nontrivial order-two subgroup fixing a, whereas a transitive degree-six action of this order-six group would be regular. This final contradiction proves the proposition. □

Source and boundary. This replaces the transitive-side equality-graph step in Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 5.26, physical page 34. The complete numerical transitive-side proof, unique-crossing lemma, paired-row parity lemma and retained Whiston bounds are paid inputs. The binary-kernel argument is written in full; no finite computation or new formal theorem is required. This note does not remove the separate intransitive-side D_{10} residual, prove every transitive-tail rank bound, or complete the whole L044 source. No original, question or priority credit is asserted.